



Introduce R to your friends

06.21
2025

R as a workflow component

Hanna Winter

About Geosyntec

Founded **1984**

Employees **2,600+ worldwide**

Headquarters **Boca Raton, FL**

NW Offices

Bainbridge Island, WA

Bellingham, WA

Olympia, WA

Seattle, WA

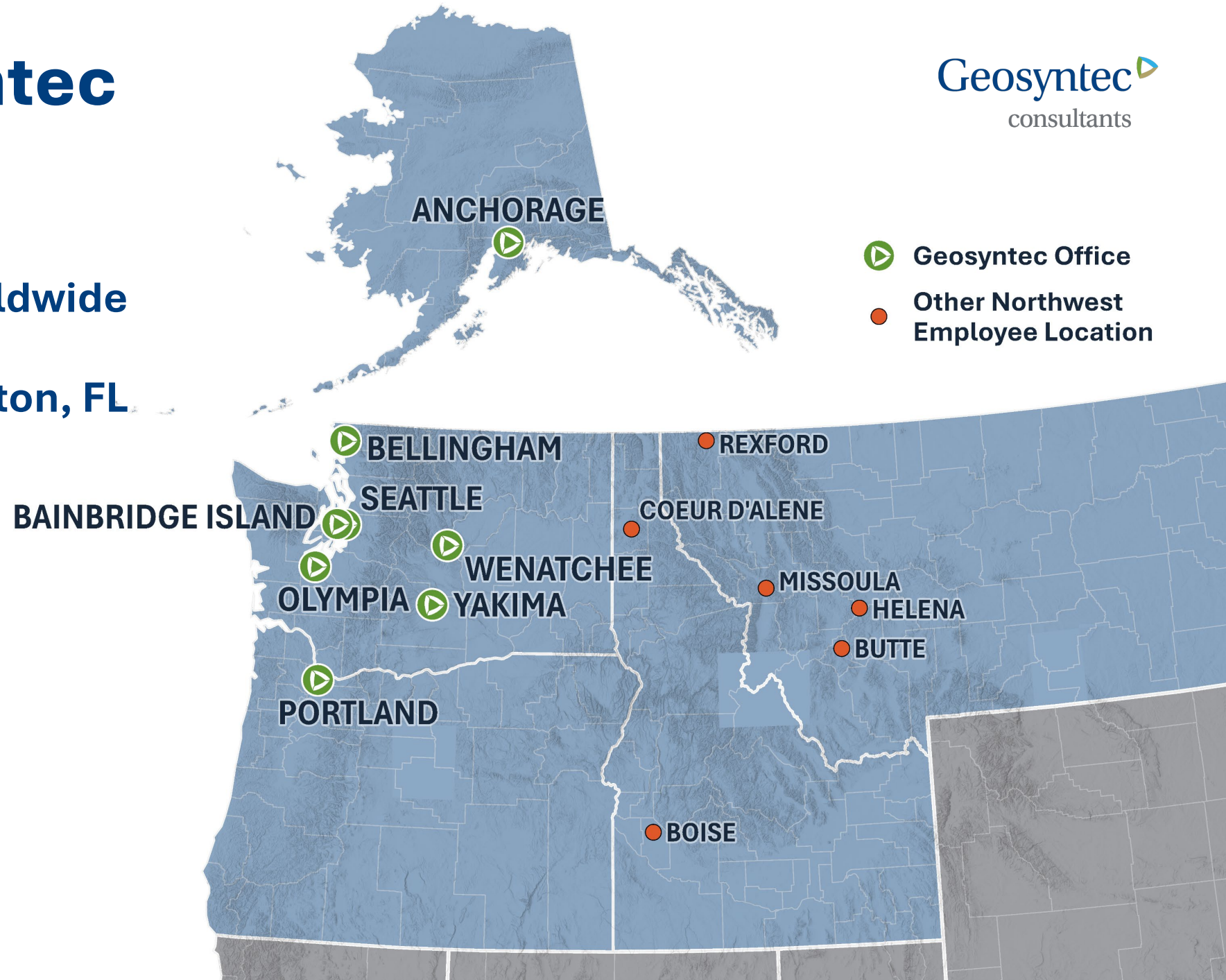
Wenatchee, WA

Yakima, WA

Portland, OR

Anchorage, AK

Geosyntec
consultants



Practice Areas



INFRASTRUCTURE

Geotechnical
Engineering

Geologic Evaluation
and Site Hazard
Assessment



ENVIRONMENTAL

Site Assessment,
Remediation, and
Redevelopment

Solid Waste



STORMWATER

Municipal and
Industrial Stormwater
Planning, Design, and
Permitting

Stormwater and
Surface Water
Monitoring



WATER RESOURCES

Water Rights and
Marketing, Supply,
Storage, and Planning



DATA + MAPPING

GIS, Drafting, Data
Management,
Development,
Visualization,
Technology
Implementation, and
Planning

Presenter



Hanna Winter Environmental Data Scientist

- BS + MS Environmental Science
- 15 years experience in natural resources management
- Water quality, regulatory compliance
- Database management and environmental data science
- I love to streamline workflows!

Bellingham, WA



R in Consulting

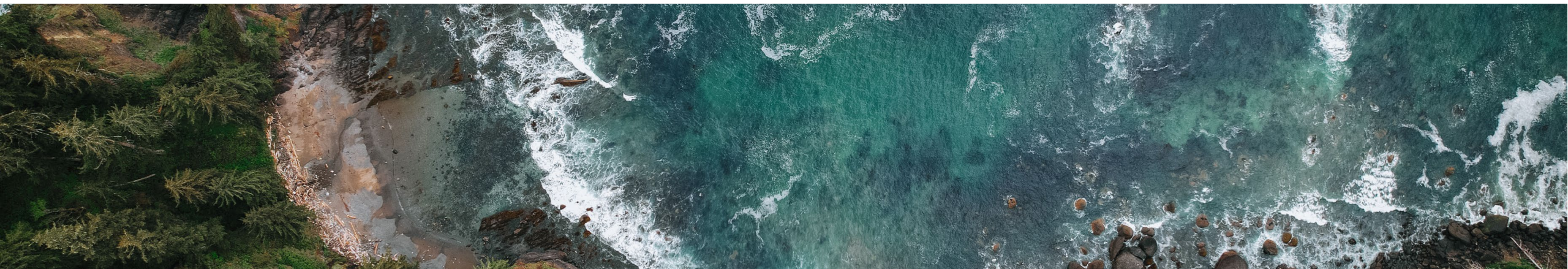
- Platform agnostic
- Right tool for the job
- No task too small for R
- Internal advertising

Examples
No task is too small



Compile or combine files

-  Weekly downloads
-  Merge related tables
-  Metadata and header info
-  QC Sheets
-  Excel outputs



Automate the Routine

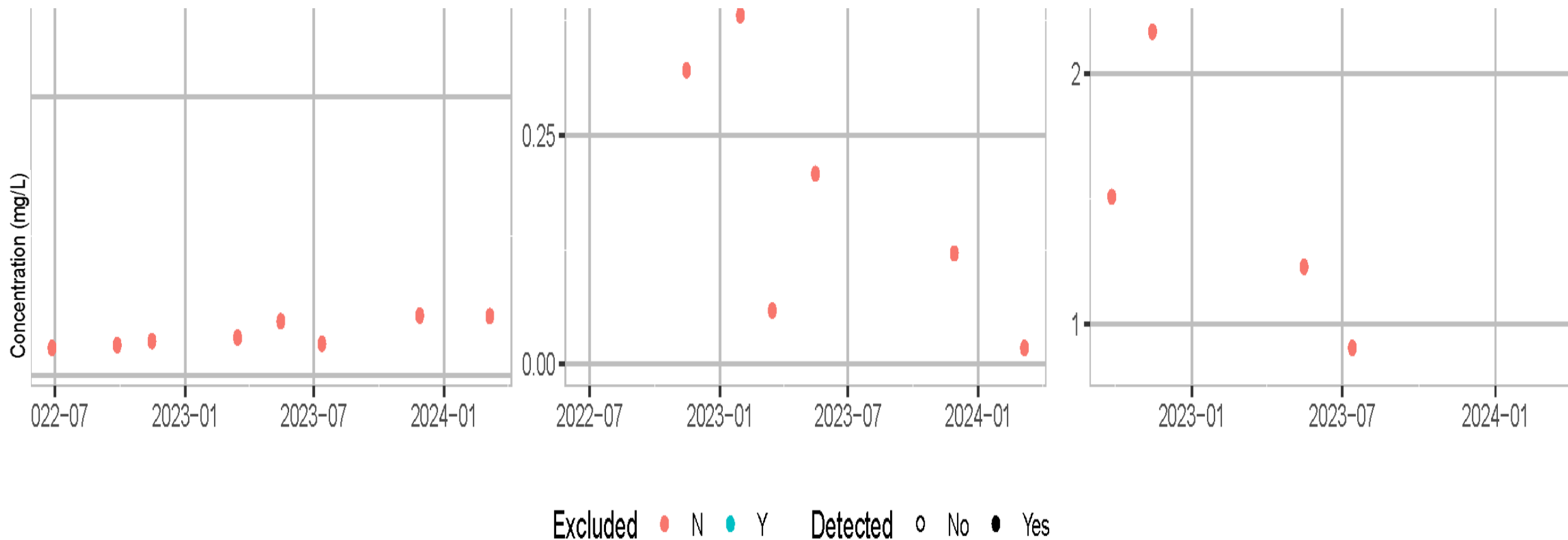
- Wrangle
- QC
- Calculations
- Reformat
- Multiple outputs



Plots!



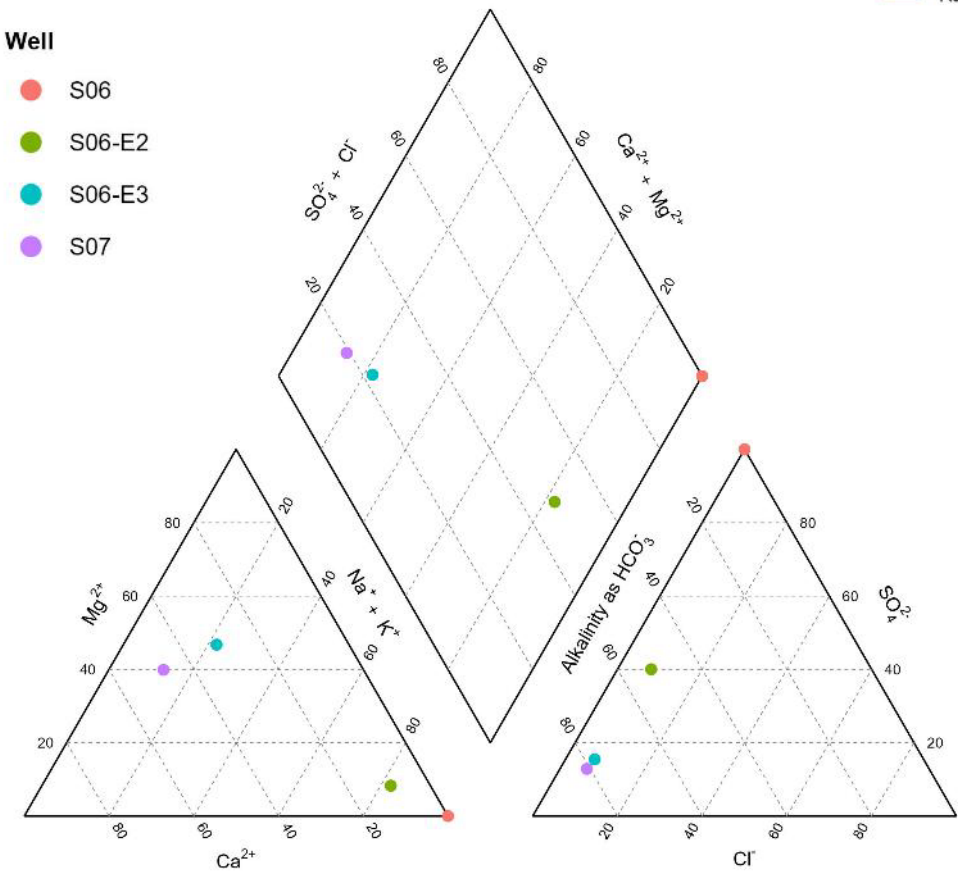
Data visualization as QC



**But this one is
complicated and
requires specialty
software**

That's ok!

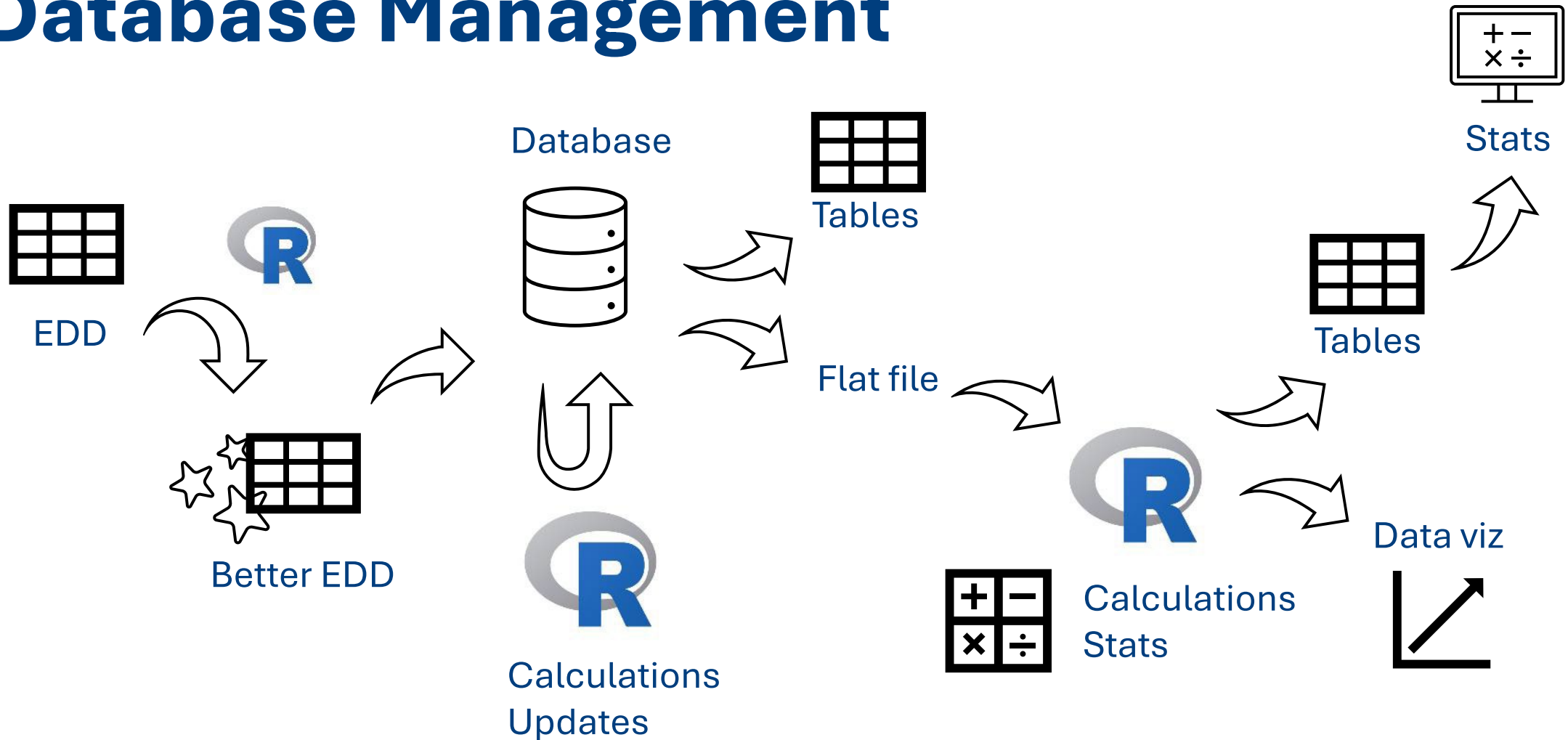
Piper Plot - mean concentration



Workflows



Database Management

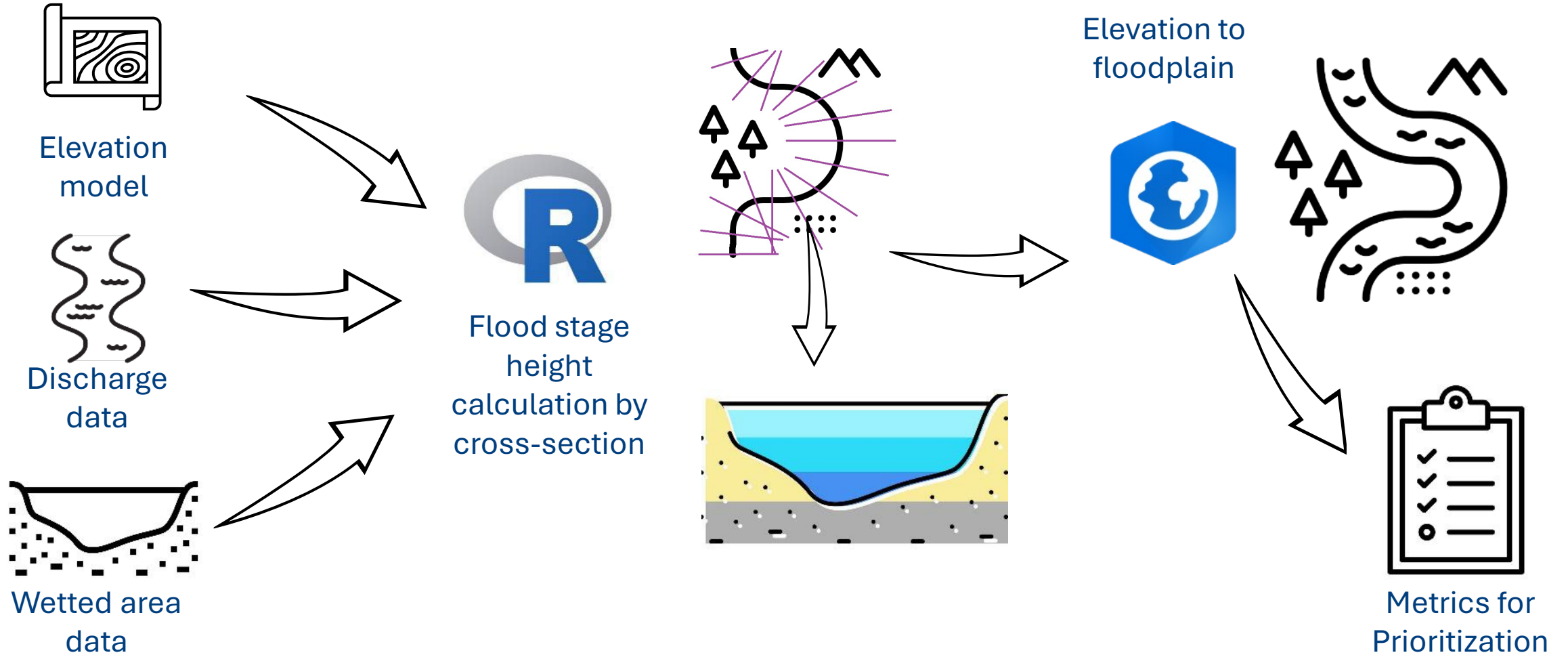




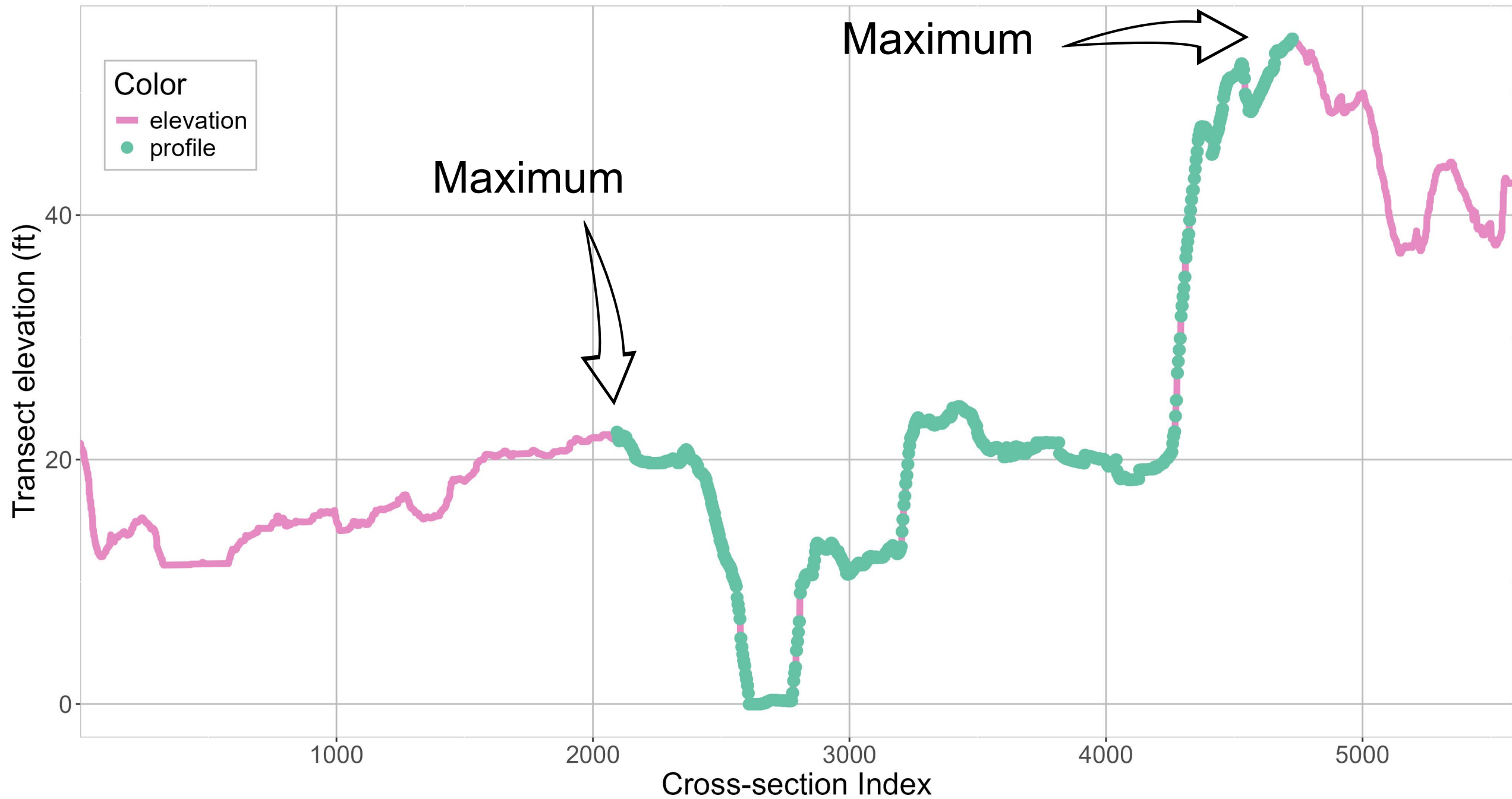
Upper Columbia Salmon Recovery Board

Floodplain delineation

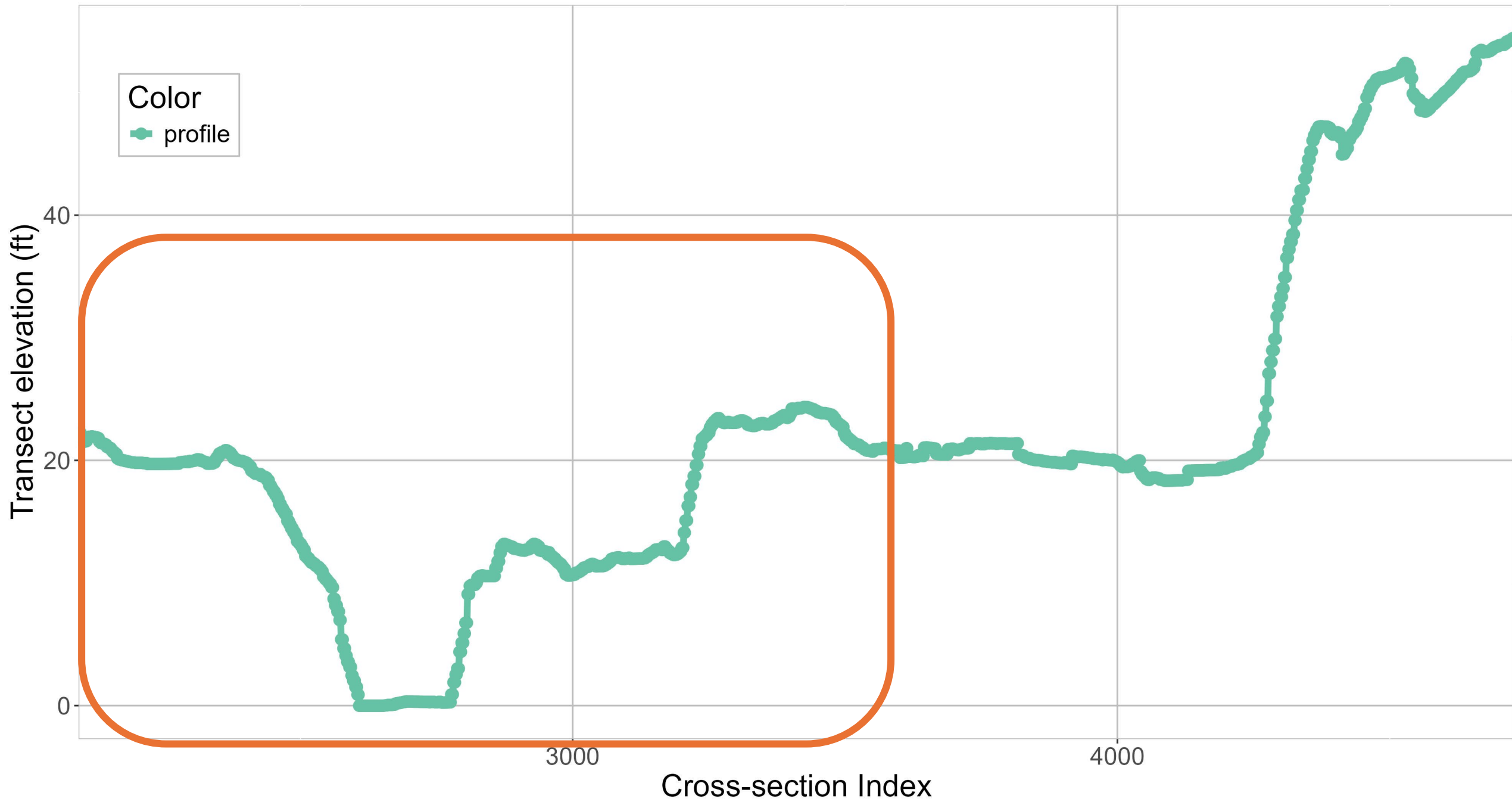
Data pipeline



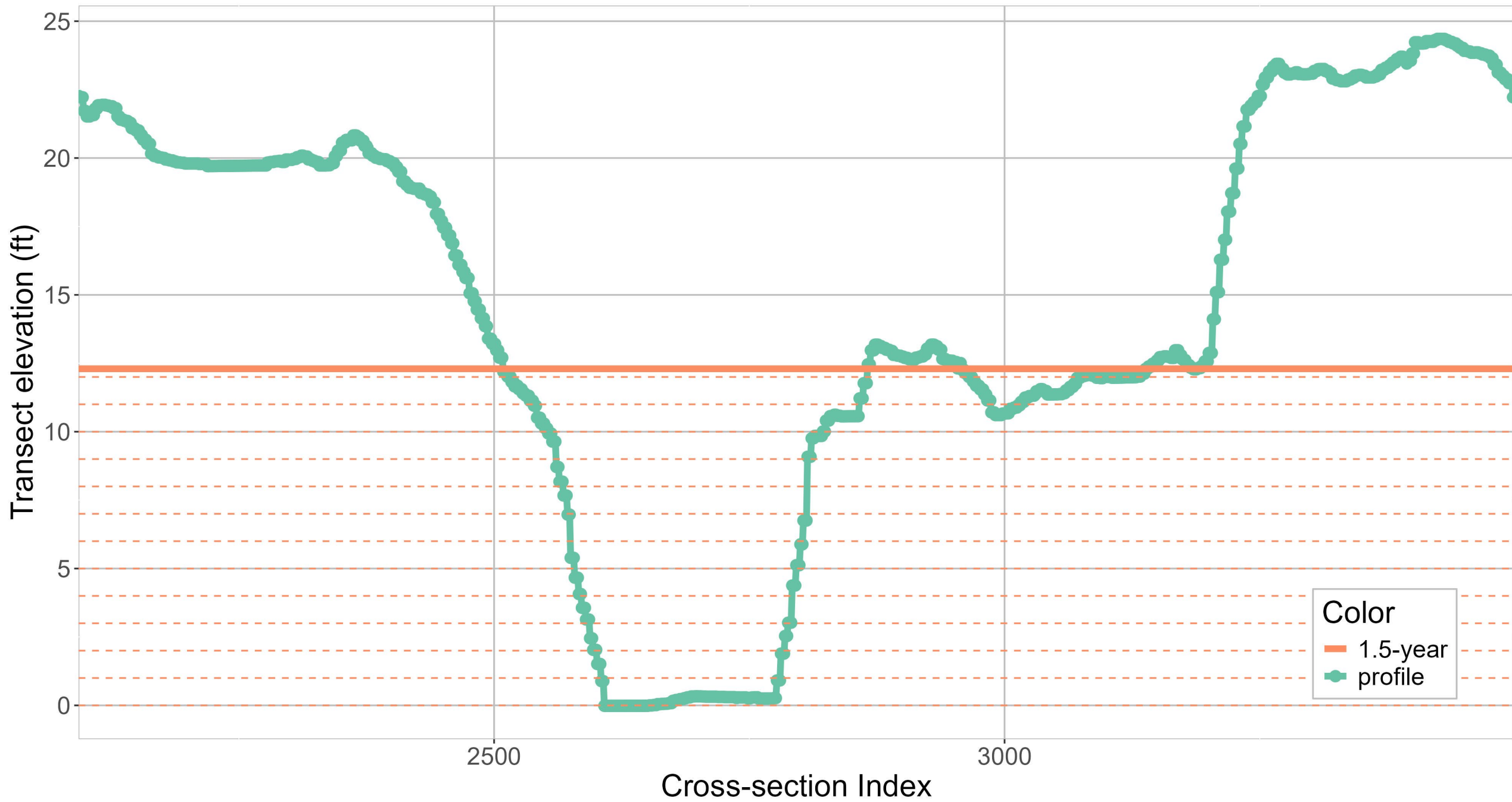
Wenatchee River Cross-Section



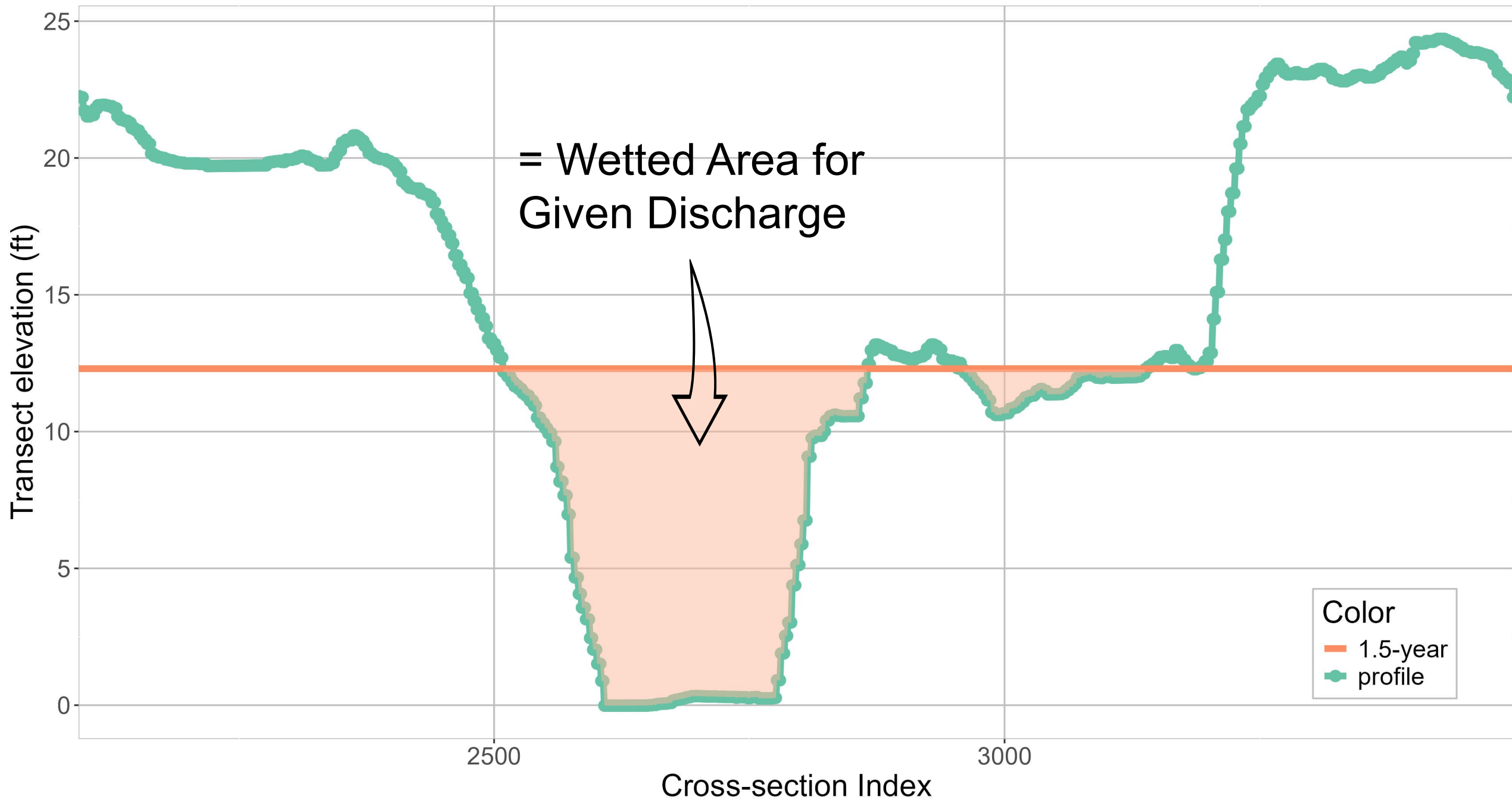
Wenatchee River Cross-Section



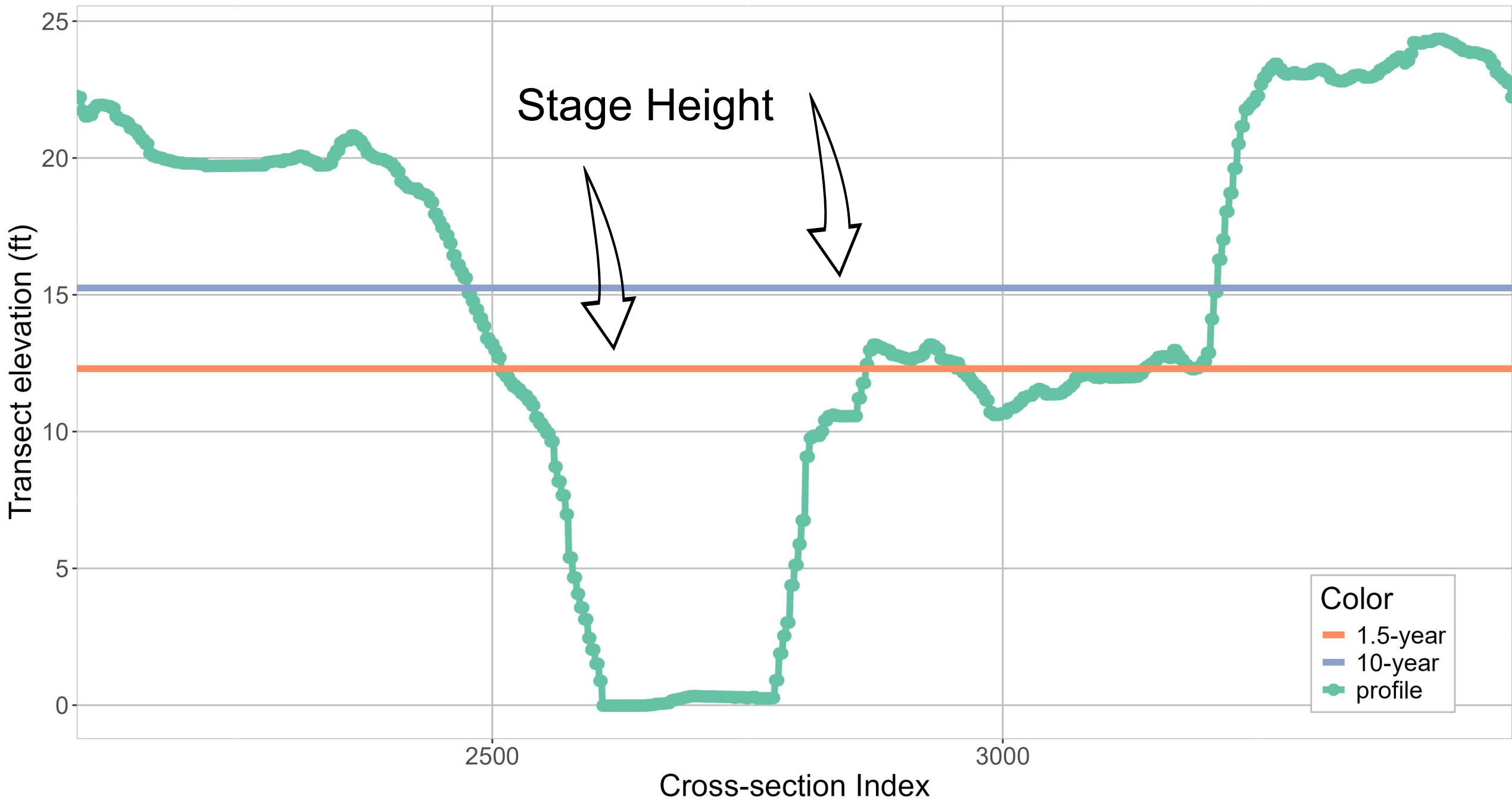
Wenatchee River Cross-Section



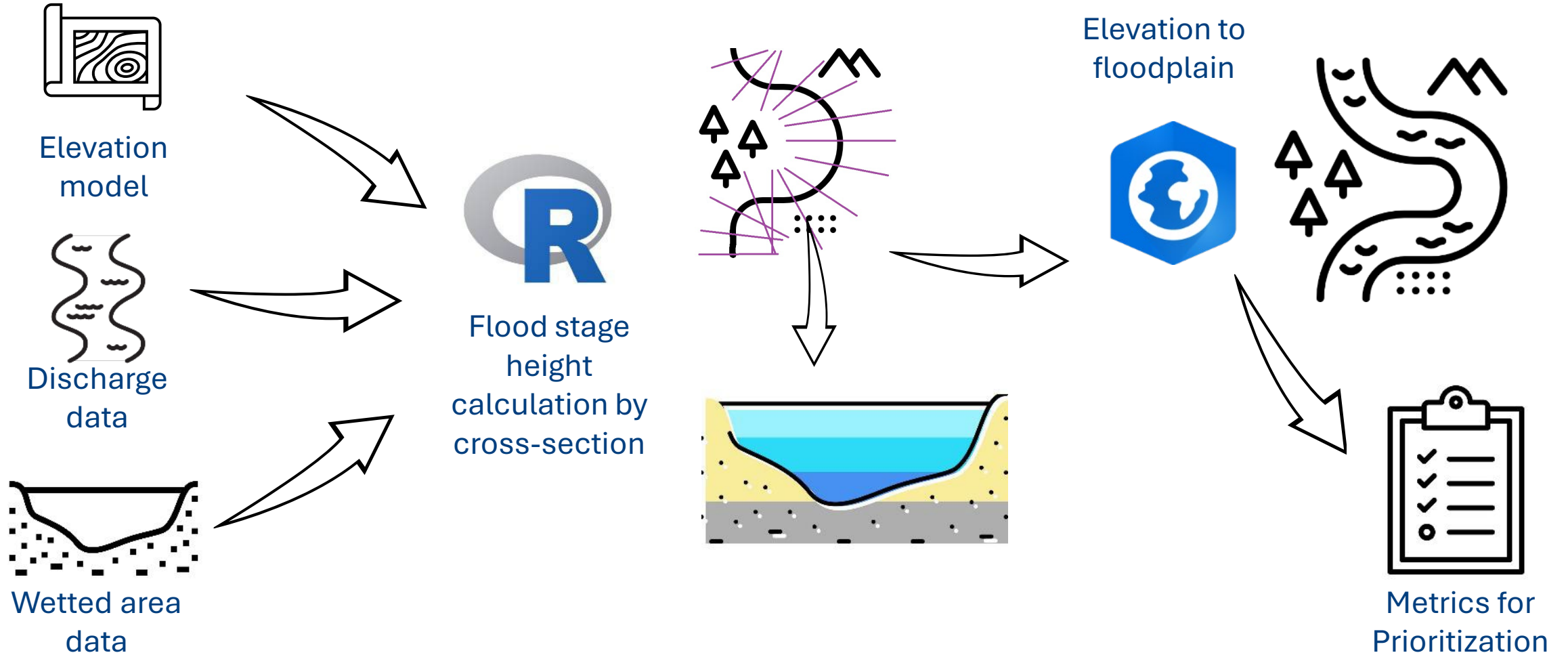
Wenatchee River Cross-Section



Wenatchee River Cross-Section

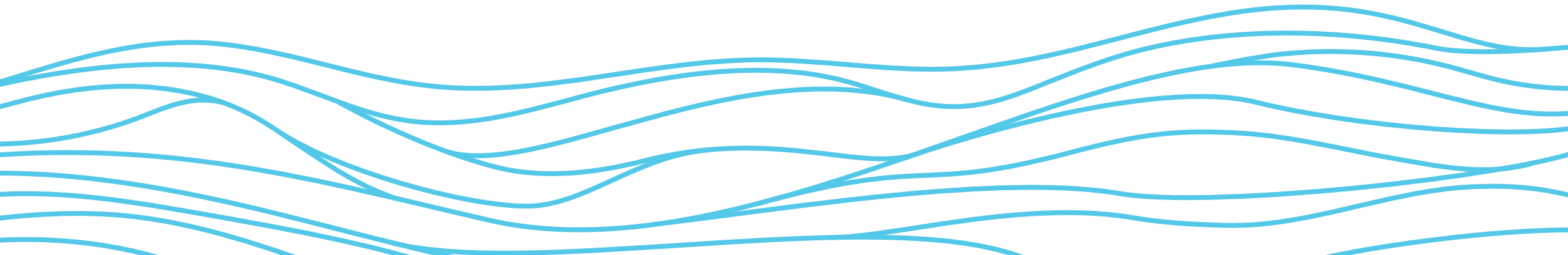


Data pipeline



Methow Okanogan Beaver Project

Streamflow restoration



Methow Beaver Project

Transducer timeseries data

When

- 15-minute interval time-series data

+

- Quarterly Downloads

+

- 2-3 years of data

+

- 55 monitoring stations

Needs lots of processing

Compensate pressure

Pair with manual measurements

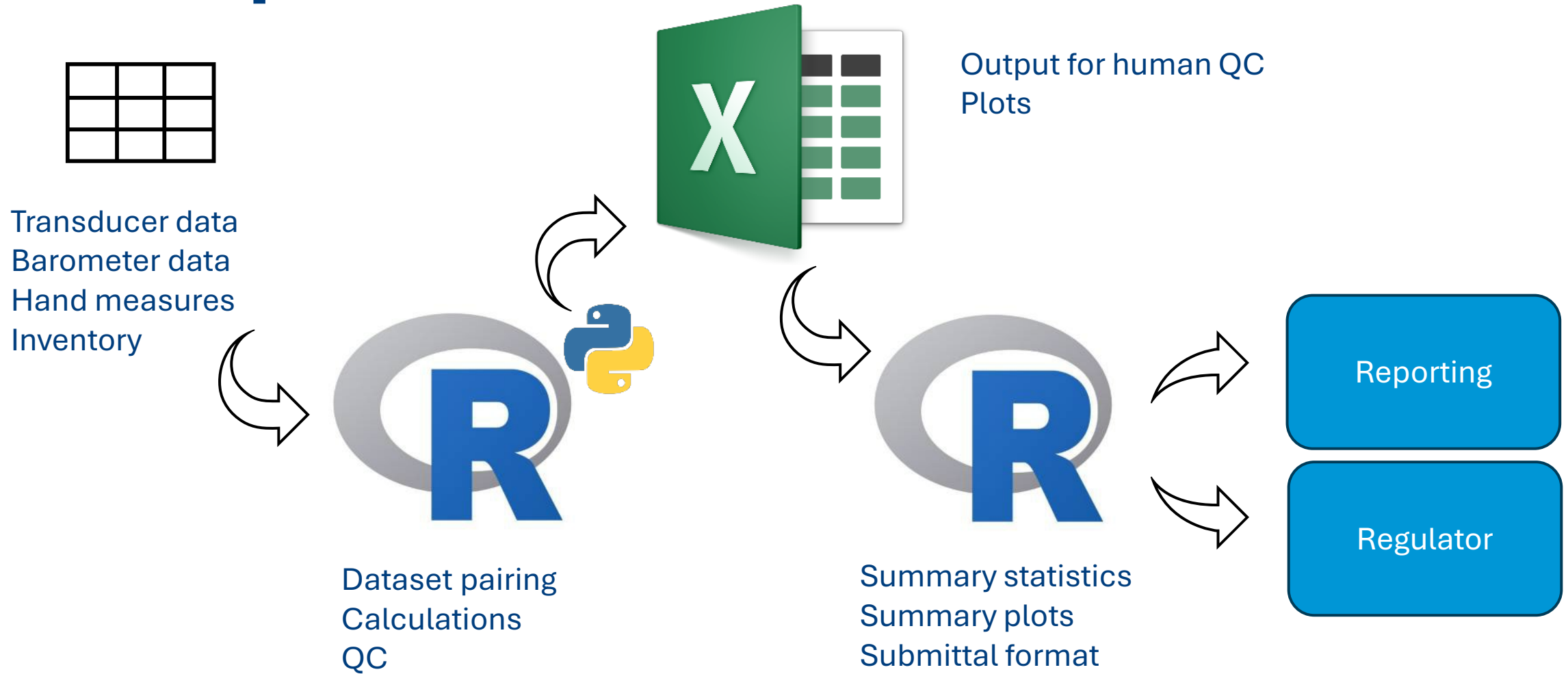
Calculations

QC

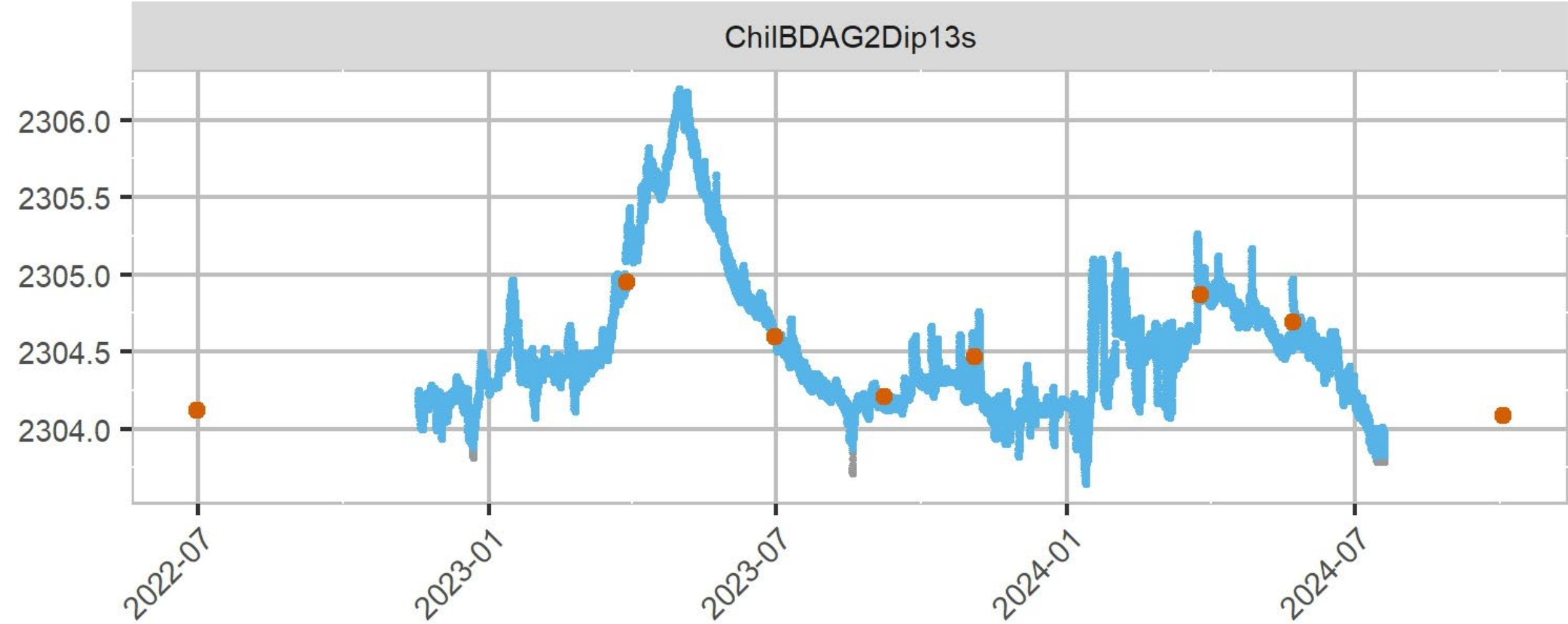
Submit to regulator



Data Pipeline



Hydrographs



Questions/Thank you

Hanna Winter

Environmental Data Scientist

Hanna.Winter@geosyntec.com



geosyntec.com